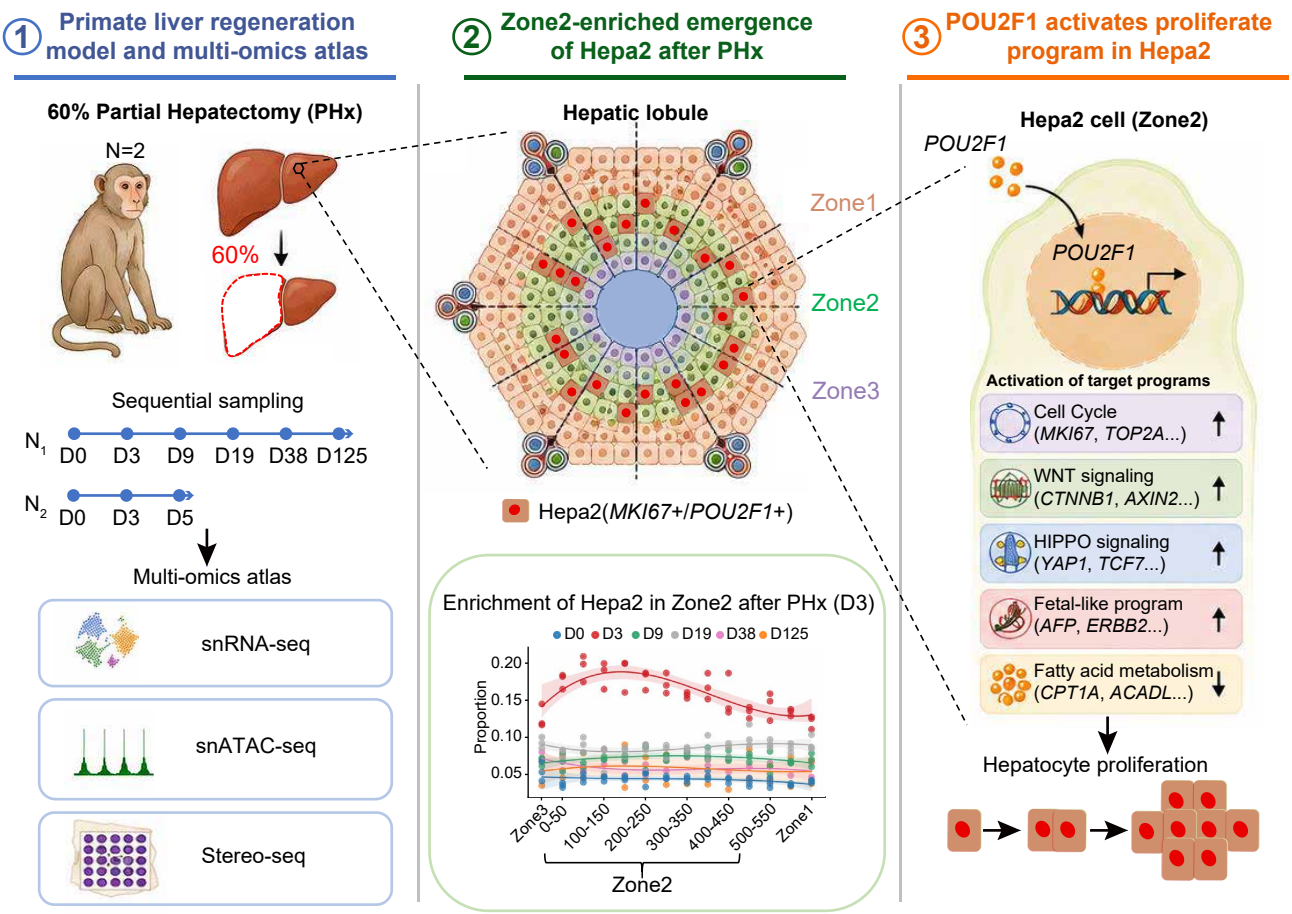

① Primate liver regeneration model and multi-omics atlas
60% Partial Hepatectomy (PHx)
N=2
60%
Sequential sampling
N1 D0 D3 D9 D19 D38 D125
N2 D0 D3 D5
Multi-omics atlas
snRNA-seq
snATAC-seq
Stereo-seq
② Zone2-enriched emergence of Hepa2 after PHx
Hepatic lobule
Zone1
Zone2
Zone3
Hepa2(MKI67+/POU2F1+)
Enrichment of Hepa2 in Zone2 after PHx (D3)
D0 D3 D9 D19 D38 D125
Proportion
0.20
0.15
0.10
0.05
Zone3 0-50 100-150 200-250 300-350 400-450 500-550 Zone1
Zone2
③ POU2F1 activates proliferate program in Hepa2
Hepa2 cell (Zone2)
POU2F1
POU2F1
Activation of target programs
Cell Cycle (MKI67, TOP2A...)
WNT signaling (CTNNB1, AXIN2...)
HIPPO signaling (YAP1, TCF7...)
Fetal-like program (AFP, ERBB2...)
Fatty acid metabolism (CPT1A, ACADL...)
Hepatocyte proliferation